

Automated Trading Point Forecasting Based on Bicluster Mining and Fuzzy Inference

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Abstract—Historical financial data are frequently used in technical analysis to identify patterns that can be exploited to achieve trading profits. Although technical analysis using a variety of technical indicators has proven to be useful for the prediction of price trends, it is difficult to use them to formulate trading rules that could be used in an automatic trading system due to the vague nature of the rules. Moreover, it is challenging to determine a specified combination of technical indicators that can be used to detect good trading points and trading rules since different stock may be affected by different set of factors. In this paper, we propose a novel trading point forecasting framework that incorporates a bicluster mining technique to discover significant trading patterns, a method to establish the fuzzy rule base, and a fuzzy inference system optimized for trading point prediction. The proposed method (called BM-FM) was tested on several historical stock datasets and the average performance was compared with the conventional buy-and-hold strategy and five previously reported intelligent trading systems. Experimental results demonstrated the superior performance of the proposed trading system.

Index Terms—Biclustering, fuzzy inference system, particle swarm optimization, technical analysis, trading point prediction, trading rules.

I. INTRODUCTION

STOCK price forecasting has always been of interest to investors and professional analysts. Success in stock trading

depends on picking the best time to buy or sell stocks. However, accurate stock price forecasting is very difficult due to the inherently noisy environment and the high volatility of price movement [1]. Stock price movement is related to a number of factors including company performance, government policies, economic environments, and inflation rates. Technical analysis, which analyzes the price movement of a stock through charts and mathematical formulas called technical indicators in order to identify patterns that can be exploited for profitable trading, is a common approach for stock market prediction in financial engineering [2], [3]. Despite the advances in technical analysis, it is still difficult to forecast accurately stock price and trading points in financial markets.

With the rapid development of machine learning techniques in recent years, a large body of algorithms has been proposed to provide decision support to investors and professional analysis. These techniques are often used in stock trading in two ways: stock price prediction [4]–[12] and stock trading point prediction [13]–[22].

In stock price prediction, Huang and Tsai [8] applied support vector regression and self-organizing feature map to forecast the stock price. Hsieh *et al.* [9] integrated Haar wavelet, artificial bee colony algorithm, stepwise regression-correlation selection, and recurrent neural networks together to predict the stock index price. Svalina *et al.* [10] applied an adaptive network-based fuzzy inference system (ANFIS) to predict the stock price. Laboissiere *et al.* [11] used artificial neural networks (ANN) to model the relationship between the stock price and the input features. Chen and Chen [12] applied fuzzy time series and fuzzy variation groups to predict the daily Taiwan Stock Exchange Capitalization Weighted Stock Index.

In stock trading point prediction, Bekiros [16] proposed a hybrid neuro fuzzy system for decision making. Wen *et al.* [18] proposed a new intelligent trading system based on oscillation box prediction by combining stock box theory and support vector machine (SVM) algorithm. Chang *et al.* [19] proposed a model using the piecewise linear representations and ANN to detect stock trading signals. As ANN lacks explanatory ability and has over-fitting problem, Luo and Chen [20] integrated piecewise linear representation (PLR) and weighted SVM (WSVM) to forecast the stock trading signals and compared the result with the method based on PLR and ANN. Ng *et al.* [13] presented a method that is based on using a genetic algorithm (GA) to minimize a weighted localized generalization error to predict trading points. Their approach, however, requires high-computational

Manuscript received May 26, 2018; revised September 12, 2018 and December 17, 2018; accepted March 6, 2019. Date of publication March 14, 2019; date of current version February 3, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61571193, in part by the Natural Science Foundation of Guangdong Province, China under Grant 2017A030312006, and in part by the Science and Technology Program of Guangzhou under Grant 201704020134. (Corresponding authors: Alan Wee-Chung Liew and Qinghua Huang.)

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This paper has supplementary downloadable material available at <http://ieeexplore.ieee.org>, provided by the author. The material consists of a PDF, readable with Adobe PDF reader. The size of the file is 360 KB. Contact a.liew@griffith.edu.au for further questions about this paper.

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Digital Object Identifier 10.1109/TFUZZ.2019.2904920

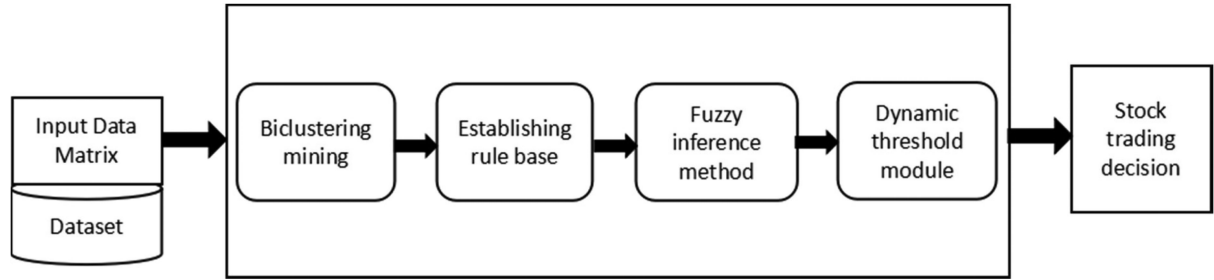


Fig. 1. Flowchart of the proposed BM-FM algorithm.

time for the GA to converge. Considering that the behavioral bias of human will affect the stock price, Caillec *et al.* [23] presented possibilistic and probabilistic method combining several technical indicators and fusion operators to make trading decision.

In the studies mentioned above, a variety of technical indicators were often used as input features in the forecast models. As learning technical indicators jointly could provide better trading information about the future trends of stock price than learning each indicator independently [21], [24], [25]. Huang *et al.* [26] proposed a novel method integrating biclustering algorithm and the K -nearest neighbor (BIC-K-NN) to determine the trading actions. However, the use of a unified distance between the trading day and the biclusters of technical indicators could sometime lead to unreliable trading points.

In this paper, we propose a novel intelligent trading system in which a biclustering algorithm is first applied to mine significant technical trading patterns from the technical indicators, then, a fuzzy inference system is used to generate the trading actions. As the proposed system makes use of biclustering pattern mining and fuzzy modeling, we call it BM-FM. The biclustering algorithm extracts local coherent patterns (regarded as trading patterns), each of which contains a subset of technical indicators and a subset of trading days. As a result, a combination of different technical indicators having a coherent pattern can be found if they behave in a similar manner for the same sort of turning points. Each bicluster can be used to form a fuzzy rule, which contains information about the possible future trends of stock price. The fuzzy rules can then be used to establish the rule base. Finally, a fuzzy inference system is used to generate trading action, which executes the generated trading rules for an arbitrary stock.

The rest of the paper is organized as follows. Section II describes the proposed BM-FM method in detail. Section III presents and discusses the experimental results of the proposed method and the comparisons with five other trading methods. Section IV draws the conclusion on the proposed method.

II. METHODS

Fig. 1 shows the flowchart of the proposed BM-FM algorithm. A set of technical indicators with varying time period and future return for each trading day are used to form the two-dimensional (2-D) data matrix. Biclustering is applied to the matrix to

discover local coherent patterns (i.e., biclusters) in which a subset of technical indicators with constant values associated with a subset of trading days lead to similar future returns. The discovered biclusters are then used to generate the fuzzy rule base for the trading signals. A Mamdani fuzzy inference system is then used to generate the combined output trading signals. Finally, a dynamic threshold module learns the optimal threshold value by using particle swarm optimization for future trading decision making (i.e., buy or sell).

A. Biclustering Algorithm Overview

The term biclustering was first proposed by Cheng and Church [27] for mining gene expression patterns in gene expression data analysis. Unlike traditional clustering, biclustering can discover the local coherent pattern in a 2-D data matrix by clustering the data along the row and the column simultaneously [28]. Tables I and II show an example of obtaining a bicluster.

Biclusters can be grouped into four types: biclusters with constant values, constant values in rows or columns, coherent values, and coherent evolutions. A large number of successful applications exist for biclustering in many domains, such as biological data analysis, target marketing, feature selection [29]–[35]. In this paper, we focus on bicluster with constant columns (i.e., the technical indicators corresponding to the columns have the same values as illustrated in Tables I and II).

B. Data Preparation

To detect the biclusters, historical data are used to construct a data matrix, in which the rows are the trading days and the columns correspond to the technical indicators with different time periods (days), and future returns. The historical data of a stock consists of: trading date, opening price, highest price, lowest price, closing price, and transaction volume. In this study, 13 technical indicators are selected: simple moving average (SMA), relative strength index (RSI), Williams %R (%R), rate of change (ROC), commodity channel index (CCI), ease of movement (EMV), ultimate oscillator (UOS), BR, AR, K , J , D , and momentum index (MTM), all of which are used extensively in the technical analysis of the stock market. Our algorithm chooses these technical indicators and different time periods to generate the trading rules. The technical indicators we used, together with the commonly used time period parameters, are presented in Table III.

TABLE I
DATA MATRIX

0.853	0.256	0.852	0.652	0.685	0.584	0.125	0.125	0.652	0.452	0.221
0.423	0.663	0.566	0.265	0.325	0.741	0.453	0.832	0.123	0.233	0.785
0.632	0.325	0.625	0.785	0.847	0.962	0.642	0.556	0.525	0.652	0.215
0.552	0.653	0.562	0.565	0.356	0.523	0.432	0.842	0.335	0.235	0.685
0.235	0.852	0.421	0.454	0.652	0.652	0.742	0.212	0.632	0.744	0.366
0.365	0.653	0.542	0.125	0.345	0.321	0.982	0.832	0.722	0.255	0.214
0.215	0.755	0.321	0.952	0.754	0.652	0.212	0.552	0.982	0.562	0.562
0.562	0.642	0.532	0.562	0.369	0.545	0.542	0.886	0.320	0.263	0.322
0.656	0.423	0.725	0.412	0.645	0.965	0.632	0.785	0.245	0.412	0.922
0.365	0.635	0.544	0.523	0.354	0.741	0.412	0.852	0.452	0.256	0.215
0.456	0.101	0.210	0.652	0.646	0.652	0.852	0.632	0.785	0.922	0.551

TABLE II
BICLUSTER OBTAINED FROM TABLE I

0.663	0.566	0.325	0.832	0.233
0.653	0.562	0.356	0.842	0.235
0.653	0.542	0.345	0.832	0.255
0.642	0.532	0.369	0.886	0.263
0.635	0.544	0.354	0.852	0.256

TABLE III
TECHNICAL INDICATORS USED AS INPUT VARIABLES

Indicators	Description
SMA	It computes the average price of a stock over a number of days. Periods used = {6, 10, 12, 24, 30}
RSI	It measures the speed and change of price movements. Periods used = {6, 12, 18, 24, 30}
%R	It reflects the level of the closing price relative to the highest high over the maximum trading range for the look-back period. Periods used = {6, 14, 20}
ROC	It measures the percent change in price from one period to the next. Periods used = {6, 12, 24, 28}
CCI	It is used to identify a new trend or to warn of extreme conditions. Periods used = {6, 12, 14, 28}
EMV	It measures the amount of volume required to move prices. Periods used = {6, 12, 14, 28}
UOS	It captures price momentum across three different time frames. Periods used = {7, 14, 28}
BR, AR	They reflect the stock's popularity. Period used = {26}
K, D, J	They are used in short-term trend analysis. Period used = {9}
MTM	It reflects short term price volatility. Period used = {6}

The formulas for the technical indicators are as follows.

1) SMA:

$$SMA(n) = \frac{\sum_{i=1}^n Cl_i}{n} \quad (1)$$

where Cl_i is the closing price at the i th trading day, n is the time period parameters.

2) RSI:

$$RS(n) = \frac{AG(n)}{AL(n)} \quad (2)$$

$$RSI(n) = 100 - \frac{100}{1 + RS(n)} \quad (3)$$

where $AG(n)$ represents the average gain over the past n days, $AL(n)$ represents the average loss over the past n days.

3) ROC:

$$ROC(n) = \frac{Cl_i - Cl_{(i-n)}}{Cl_{(i-n)}} \quad (4)$$

where Cl_i is the closing price at the i th trading day, $Cl_{(i-n)}$ represents the closing price at n days before the i th trading day.

4) CCI:

$$CCI(n) = \frac{TP - SMA(n)}{0.015 \times MD(n)} \quad (5)$$

where TP presents the average of the highest price, the lowest price, and the closing price. $MD(n)$ represents the mean deviation of the past n days.

5) EMV:

$$A_i = (H_i + L_i) / 2 \quad (6)$$

$$B_i = (H_{i-1} + L_{i-1}) / 2 \quad (7)$$

$$C_i = (H_i - L_i) / 2 \quad (8)$$

$$EMV(n) = \frac{1}{n} \sum_{i=1}^n \frac{(A_i - B_i) \times C_i}{V_i} \quad (9)$$

where H_i is the highest price at the i th trading day, L_i is the lowest price at the i th trading day, V_i is the trading volume at the i th trading day.

6) %R:

$$\%R(n) = \frac{H(n) - Cl_i}{H(n) - L(n)} \times 100 \quad (10)$$

where $H(n)$ is the highest price among the past n trading days, $L(n)$ is the lowest price among the past n trading days, Cl_i is the closing price at the i th trading day.

7) UOS:

$$TH_i = \max(H_i, Cl_{i-1}) \quad (11)$$

$$TL_i = \min(L_i, Cl_{i-1}) \quad (12)$$

$$ACC1 = \sum_{i=1}^{n_1} (Cl_i - TL_i) / \sum_{i=1}^{n_1} (TH_i - TL_i) \quad (13)$$

$$ACC2 = \sum_{i=1}^{n_2} (Cl_i - TL_i) / \sum_{i=1}^{n_2} (TH_i - TL_i) \quad (14)$$

$$\text{ACC3} = \sum_{i=1}^{n_3} (\text{Cl}_i - \text{TL}_i) / \sum_{i=1}^{n_3} (\text{TH}_i - \text{TL}_i) \quad (15)$$

equation (16) shown at the bottom of this page.
where n_1, n_2, n_3 are set to 7, 14, 28, respectively.

8) AR, BR:

$$\text{AR}(n) = \frac{\sum_{i=1}^n H_i - O_i}{\sum_{i=1}^n O_i - L_i} \times 100 \quad (17)$$

$$\text{BR}(n) = \frac{\sum_{i=1}^n H_i - \text{Cl}_i}{\sum_{i=1}^n \text{Cl}_i - L_i} \times 100 \quad (18)$$

where O_i represents the opening price at the i th trading day.

9) K, D, J :

$$\text{RSV}_i = \frac{\text{Cl}_i - L(n)}{H(n) - L(n)} \times 100 \quad (19)$$

$$K_i = (1 - \alpha) \times K_{i-1} + \alpha \times \text{RSV}_i \quad (20)$$

$$D_i = (1 - \beta) \times D_{i-1} + \beta \times K_i \quad (21)$$

$$J_i = 3 \times D_i - 2 \times K_i \quad (22)$$

where α and β are equal to 1/3. K_i, D_i, J_i denote the value of indicators K, D, J at the i th trading day, respectively.

10) MTM:

$$\text{MTM}(n) = \frac{\text{Cl}_i}{100 \times \text{Cl}_{(i-n)}} - 100 \quad (23)$$

where Cl_i is the closing price at the i th trading day, $\text{Cl}_{(i-n)}$ represents the closing price of n days before the i th trading day.

In order to capture the short term movement direction of stock price, the future return rate (FRR) is calculated for every trading day of a stock. To compute FRR, we compute the smoothed closing price of each trading day by the mean closing price of the next N trading days (denoted by ACP_i) as

$$\text{ACP}_i = \frac{1}{N} \sum_{M=i}^{i+N-1} \text{CP}_M \quad (24)$$

where CP_i is the closing price at the i th trading day and $N = 6$. Then, the future return rate corresponding to each trading day is computed as:

$$\text{FRR}_i = \frac{\text{ACP}_i - \text{CP}_i}{\text{CP}_i} \times 100\% \quad (25)$$

where FRR_i denotes the future return rate corresponding to the i th trading day, which represents the short term movement direction of stock price. The following criteria are used to define the price trend, where T_t is empirically set to 0.5% (i.e., a daily price change of 0.5%), which is used to quantify the extent of future return. This value is normally set based on experience or can be optimized through grid search.

Upward trend:

$$\text{if } T_t \leq \text{FRR}_i < 2T_t, \text{ set } \text{FRR}_i = 1$$

$$\text{if } 2T_t \leq \text{FRR}_i < 3T_t, \text{ set } \text{FRR}_i = 2$$

$$\text{if } \text{FRR}_i \geq 3T_t, \text{ set } \text{FRR}_i = 3.$$

Down trend:

$$\text{if } -2T_t \leq \text{FRR}_i < -T_t, \text{ set } \text{FRR}_i = -1$$

$$\text{if } -3T_t \leq \text{FRR}_i < -2T_t, \text{ set } \text{FRR}_i = -2$$

$$\text{if } \text{FRR}_i \leq -3T_t, \text{ set } \text{FRR}_i = -3.$$

Sideway trend:

$$\text{otherwise, set } \text{FRR}_i = 0.$$

As a result, we obtain the data matrix as illustrated in Table IV, consisting of 34 technical indicators and the FRR. We use Indicator Name_Period to denote the technical indicator with a specific time period (e.g., SMA_6 represents the indicator SMA with a 6-day period).

As the values of different technical indicators can vary greatly, a data normalization process is applied to each column to normalize the data into the range of [0, 1]. The data normalization applied to each column is given by [31]:

$$v_n(i, j) = \frac{v_o(i, j) - v_{\min}(j)}{v_{\max}(j) - v_{\min}(j)} \quad (26)$$

where $v_o(i, j)$ denotes the original value of the j th technical indicator at the i th trading day in the data matrix, $v_{\max}(j)$ and $v_{\min}(j)$ denote the theoretical maximum and minimum values of the j th technical indicator, respectively, and $v_n(i, j)$ denotes the normalized value of the j th technical indicator at the i th trading day.

C. Bicluster Mining

A bicluster is a subset of technical indicators, which express similar local coherent pattern. Different trading rules can be obtained from different combination of technical indicators, where each indicator takes similar value over a subset of rows in a discovered trading pattern. We utilize biclustering algorithm to find biclusters with constant columns from the historical data. The procedure is shown in Fig. 2. The bicluster mining consists of the following steps.

- 1) *Step 1*: For each column of the technical indicators, an agglomerative hierarchical clustering method [36] using the average linkage with a distance threshold T_{hc} is applied to measure the similarity of elements. The detected clusters are treated as bicluster seeds.
- 2) *Step 2*: For a bicluster seed, a new column can be added to it if the newly formed submatrix satisfies the mean squared residue (MSR) score [26] (i.e., the MSR score of the bicluster is less than a preset threshold T_{msr}). This procedure is repeated until no new column can be added to the submatrix. The submatrix is then output as a bicluster.

$$\text{UOS} = \frac{\text{ACC1} \times n_2 \times n_3 + \text{ACC2} \times n_1 \times n_3 + \text{ACC3} \times n_1 \times n_2}{n_1 \times n_2 + n_1 \times n_3 + n_2 \times n_3} \quad (16)$$

TABLE IV
MATRIX OF TECHNICAL INDICATORS AND FUTURE RETURN RATE FOR EACH TRADING DAY

Date	SMA_6	SMA_10		J_9	MTM_6	FRR
2000/05/10	0.652	0.821		0.792	0.345	3
2000/05/10	0.591	0.642		0.254	0.365	2
... ..	0.456	0.422		0.366	0.654	2
	0.663	0.359		0.456	0.478	-3
	0.433	0.752		0.852	0.214	-2
2006/06/10	0.551	0.656		0.541	0.662	-1
2006/06/11	0.253	0.565		0.633	0.452	2

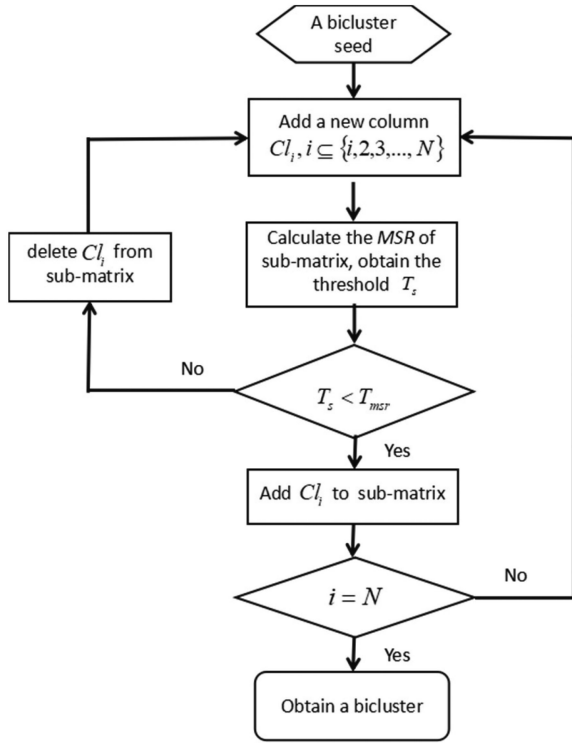


Fig. 2. Search procedure in bicluster mining.

- 3) *Step 3*: When there are two bicluster seeds constructed by the same subset of rows but at different columns, the expansion procedure may produce the same bicluster. In this case, duplicated biclusters are filtered out, keeping only one bicluster.

The type of trading pattern of a bicluster is determined by the column of future returns. Note that a valid bicluster should contain at least a minimum number of rows (set to be five) in order to be used as a robust trading rule.

D. Proposed Fuzzy Inference Method

Fuzzy systems are systems with input, output, and state variables defined on the fuzzy set. Trading heuristics based on the technical indicators are often imprecise and descriptive. For example, a trader may use the following heuristic.

“A buy (sell) signal is generated if the short term moving average crosses a longer term moving average from below (above). Usually, the larger the difference between the two averages, the

stronger the buy (sell) signal. However, if this difference is too big, the stock may be overbought (oversold) and a small sell (buy) order should be placed to protect the investment.”

Fuzzy logic allows one to deal naturally with this imprecision by using linguistic variables (such as big, medium, small) defined over a fuzzy membership function. However, traditional fuzzy inference system obtains fuzzy rules by expert’s experience, which is often difficult to obtain. Moreover, fuzzy rules obtained using expert experience are often very subjective and can differ between experts. In contrast, our method uses the biclustering algorithm to mine the fuzzy trading rules automatically from the historical stock trading database.

The fuzzy inference procedure is shown in Fig. 3. As the detected biclusters provide useful trading patterns, each bicluster is used to construct a fuzzy rule. Let us suppose that a bicluster detected from a training set contains m days and n technical indicators and the technical indicators are represented as $I_1, I_2, \dots, I_n$, respectively. The average value of the technical indicator in each column is computed as

$$I_{av_j} = \frac{1}{m} \sum_{i=1}^m v_{ij} \quad (27)$$

where v_{ij} denotes the value of the j th technical indicator I_j at the i th trading day in the bicluster, I_{av_j} represents the average value of I_j among m days. The average of the FRR for the bicluster is also computed. Therefore, each bicluster produces a mean vector $\mathbf{x}$ denoted as $\mathbf{x} = [I_{av_1}, I_{av_2}, \dots, I_{av_n}, \text{FRR}_{av}]$.

Next, we derive the corresponding fuzzy rule. A fuzzy rule is made up of a precondition and a postcondition. The values of the technical indicators ($I_{av_1}, I_{av_2}, \dots, I_{av_n}$) are used to constitute the precondition, and the values of FRR_{av} are used to constitute the postcondition. In this paper, we define four different degrees of membership for the value of the technical indicators: small, medium, big, and large. The bell shape functions are used to constitute the membership function as

$$\mu_p(\mathbf{x}) = e^{-\frac{(x-w_i^p)^2}{2\delta_i^p}} \quad (28)$$

where $i = 1, 2, 3, 4$ represents small, medium, big, or large, respectively. In this paper, $w_1^p = 0, w_2^p = \frac{1}{3}, w_3^p = \frac{2}{3}, w_4^p = 1, \delta_1^p = \delta_2^p = \delta_3^p = \delta_4^p = 1/6$. Then, four membership functions $\mu_{p\text{-small}}(\mathbf{x}), \mu_{p\text{-medium}}(\mathbf{x}), \mu_{p\text{-big}}(\mathbf{x}), \mu_{p\text{-large}}(\mathbf{x})$ can be obtained and the membership function of the technical indicator is as shown in Fig. 4.

In a similar way, we define four degrees of membership for FRR, i.e., negative small (NS), negative large (NL), positive

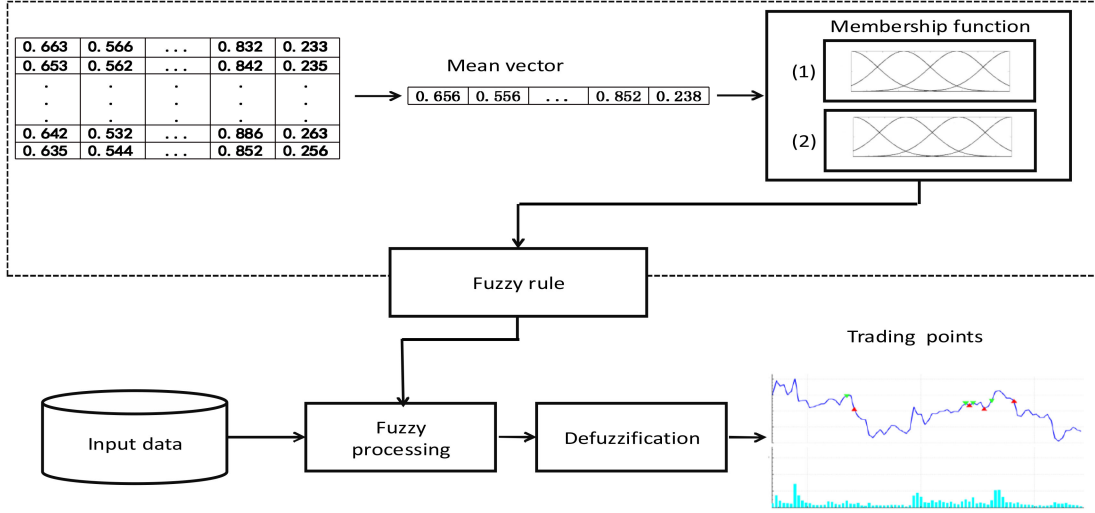


Fig. 3. Fuzzy inference procedure.

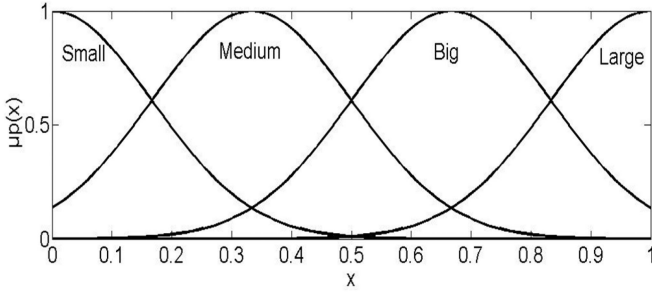


Fig. 4. Membership function of the technical indicators.

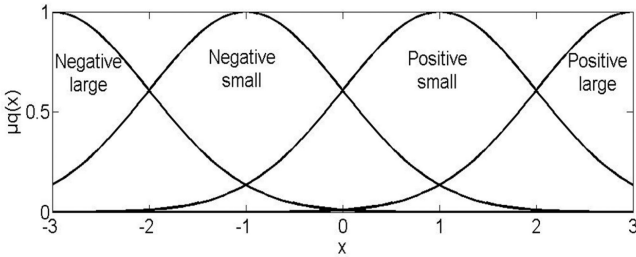


Fig. 5. Membership function of FRR.

small (PS), and positive large (PL). The PS and PL represent the possibility of stock appreciation. The NS and NL represent the possibility of stock depreciation. The FRR membership function is defined as

$$\mu_q(x) = e^{-\frac{(x-w_j^q)^2}{2\delta_j^q}} \quad (29)$$

where $j = 1, 2, 3, 4$ represent NL, NS, PS, and PL, respectively, $w_1^q = -3$, $w_2^q = -1$, $w_3^q = 1$, $w_4^q = 3$, $\delta_1^q = \delta_2^q = \delta_3^q = \delta_4^q = 1$. Then, four membership functions $\mu_{q-NL}(x)$, $\mu_{q-NS}(x)$, $\mu_{q-PS}(x)$, $\mu_{q-PL}(x)$ can be obtained and the membership function of the FRR is shown in Fig. 5.

In this paper, the MAX operation is applied to determine the membership degree of each technical indicator. The equation is defined as

$$\mu_p(I_{avj}) = \max(\mu_{p-small}(I_{avj}), \mu_{p-medium}(I_{avj}), \mu_{p-big}(I_{avj}), \mu_{p-large}(I_{avj})) \quad (30)$$

$$\mu_q(FRR_{av}) = \max(\mu_{q-NL}(FRR_{av}), \mu_{q-NS}(FRR_{av}), \mu_{q-PS}(FRR_{av}), \mu_{q-PL}(FRR_{av})) \quad (31)$$

where I_{avj} represents the average value of I_j over m days, $\mu_{p-small}(I_{avj})$, $\mu_{p-medium}(I_{avj})$, $\mu_{p-big}(I_{avj})$, $\mu_{p-large}(I_{avj})$ represent the membership degree of small, medium, big, or large, respectively. $\mu_p(I_{avj})$ denotes the membership degree of a technical indicator and decides the linguistic variables (one of small, medium, big, and large) that the technical indicator belongs to. Similarly, $\mu_q(FRR_{av})$ can be calculated and decides the linguistic variables (one of NS, NL, PS, and PL) that FRR belongs to.

Based on the above-mentioned analysis, we can evaluate the membership degree of the average technical indicators and FRR_{av} in $x = [I_{av1}, I_{av2}, \dots, I_{avn}, FRR_{av}]$ to generate the fuzzy rule. The general form of the fuzzy rule is

$$\text{Rule}^{(r)} : \text{IF } I_1 \text{ is } A_1^{(r)} \text{ and } I_2 \text{ is } A_2^{(r)}, \dots, \text{ and } I_n \text{ is } A_n^{(r)} \text{ then } FRR^{(r)} \text{ is } A_{FR}^{(r)}$$

where $\text{Rule}^{(r)}$ denotes the r th fuzzy rule, I_j denotes the j th model input, i.e., the technical indicator in the corresponding bicluster $j = 1, 2, 3, \dots, n$. $FRR^{(r)}$ represents the local inferred output of the r th fuzzy rule. $A_j^{(r)}$ denotes the precondition linguistic variables (one of small, medium, big, and large). $A_{FR}^{(r)}$ denotes the postcondition linguistic variables (one of NS, NL, PS, and PL). According to $FRR^{(r)}$, a negative value represents excellent opportunity to sell and a positive value represents excellent opportunity to buy. Once the rule base is obtained, fuzzy

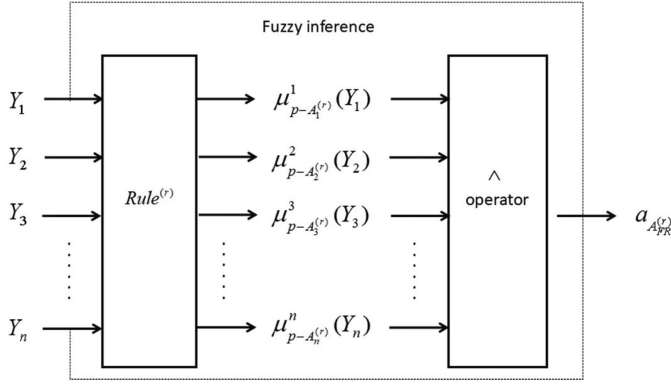


Fig. 6. Flowchart of fuzzy inference.

inference can be performed. The flowchart of fuzzy inference is shown in Fig. 6.

Mamdani fuzzy inference [37] is used in our fuzzy inference system. In this paper, r represents the r th rule in the rule set R . Y_j , $j = 1, 2, 3, \dots, n$, are the normalized technical indicators, which constitute the r th fuzzy rule $\text{Rule}^{(r)}$. For each rule, the membership grade of input variable Y_j needs to be calculated by membership function $\mu^j_{p-A_j^{(r)}}(x)$. $A_j^{(r)}$ denotes the precondition linguistic variables (one of small, medium, big, and large) in the r th fuzzy rule. Then, the strength of the output $\alpha_{A_{FR}^{(r)}}$ (the degree of truth) corresponding to the r th fuzzy rule is computed by superimposing the membership grade of each input variable $\mu^j_{p-A_j^{(r)}}(x)$. In other words, $\alpha_{A_{FR}^{(r)}}$ is the output membership grade corresponding to the r th fuzzy rule based on the input variables. Therefore, the number of $\alpha_{A_{FR}^{(r)}}$ is equal to the number of fuzzy rules in the rule base. $\alpha_{A_{FR}^{(r)}}$ is defined as

$$\alpha_{A_{FR}^{(r)}} = \mu^1_{p-A_1^{(r)}}(Y_1) \wedge \mu^2_{p-A_2^{(r)}}(Y_2) \wedge \dots \wedge \mu^n_{p-A_n^{(r)}}(Y_n) \quad (32)$$

where $A_{FR}^{(r)}$ denotes the postcondition linguistic variables (one of NS, NL, PS, and PL). It depends on the postcondition of the corresponding r th fuzzy rule. The $\wedge$ operator in our model is the MIN function, therefore

$$\alpha_{A_{FR}^{(r)}} = \min(\mu^1_{p-A_1^{(r)}}(Y_1), \mu^2_{p-A_2^{(r)}}(Y_2), \dots, \mu^n_{p-A_n^{(r)}}(Y_n)). \quad (33)$$

As described above, all fuzzy rules according to different conclusions can be divided into four classes. In other words, when the postcondition of fuzzy rule is the same, the precondition can be combined. Let us suppose that the set R denotes the entire rule base, the set O_{NS} denotes the fuzzy rules with which the postcondition is negative small, the set O_{NL} denotes the fuzzy rules with which the postcondition is negative large, the set O_{PS} denotes the fuzzy rule with which the postcondition is positive small, and the set O_{PL} denotes the fuzzy rule with which the postcondition is positive large. Therefore, the set R is defined as

$$O_{NS} \cup O_{NL} \cup O_{PS} \cup O_{PL} = R. \quad (34)$$

Let ns_i denote the fuzzy rule in set O_{NS} , $i = 1, 2, 3, \dots, C_1$. Let nl_i denote the fuzzy rule in set O_{NL} , $i = 1, 2, 3, \dots, C_2$. Let ps_i denote the fuzzy rule in set O_{PS} , $i = 1, 2, 3, \dots, C_3$. Let pl_i denote the fuzzy rule in set O_{PL} , $i = 1, 2, 3, \dots, C_4$. The membership degree of the combined rules, w_i , $i = 1, 2, 3, 4$, using the MAX operator $\vee$ is given by

$$w_1 = \alpha_{A_{FR}^{(ns_1)}} \vee \alpha_{A_{FR}^{(ns_2)}} \vee \alpha_{A_{FR}^{(ns_3)}} \vee \dots \vee \alpha_{A_{FR}^{(ns_{C_1})}} \\ = \max\left(\alpha_{A_{FR}^{(ns_1)}}, \alpha_{A_{FR}^{(ns_2)}}, \dots, \alpha_{A_{FR}^{(ns_{C_1})}}\right) \quad (35)$$

$$w_2 = \alpha_{A_{FR}^{(nl_1)}} \vee \alpha_{A_{FR}^{(nl_2)}} \vee \alpha_{A_{FR}^{(nl_3)}} \vee \dots \vee \alpha_{A_{FR}^{(nl_{C_2})}} \\ = \max(\alpha_{A_{FR}^{(nl_1)}}, \alpha_{A_{FR}^{(nl_2)}}, \dots, \alpha_{A_{FR}^{(nl_{C_2})}}) \quad (36)$$

$$w_3 = \alpha_{A_{FR}^{(ps_1)}} \vee \alpha_{A_{FR}^{(ps_2)}} \vee \alpha_{A_{FR}^{(ps_3)}} \vee \dots \vee \alpha_{A_{FR}^{(ps_{C_3})}} \\ = \max\left(\alpha_{A_{FR}^{(ps_1)}}, \alpha_{A_{FR}^{(ps_2)}}, \dots, \alpha_{A_{FR}^{(ps_{C_3})}}\right) \quad (37)$$

$$w_4 = \alpha_{A_{FR}^{(pl_1)}} \vee \alpha_{A_{FR}^{(pl_2)}} \vee \alpha_{A_{FR}^{(pl_3)}} \vee \dots \vee \alpha_{A_{FR}^{(pl_{C_4})}} \\ = \max\left(\alpha_{A_{FR}^{(pl_1)}}, \alpha_{A_{FR}^{(pl_2)}}, \dots, \alpha_{A_{FR}^{(pl_{C_4})}}\right). \quad (38)$$

After getting the membership degree of the combined rules, defuzzification is used to superpose all the valid results to produce an explicit action. The final output x_{df} of the fuzzy model is calculated by the fuzzy mean-weighted formula from each combined rule based on the Mamdani fuzzy system [37]. The fuzzy mean-weighted output is computed as

$$x_{df} = \frac{\sum_{i=1}^n w_i \times k_i}{\sum_{i=1}^n w_i} \quad (39)$$

where k_i is the center value of the FRR membership functions, i.e., $k_1 = -3$, $k_2 = -1$, $k_3 = 1$, $k_4 = 3$.

E. Dynamic Threshold Module

To generate the buy/sell decision, our fuzzy inference system learns the threshold T_α , such that when $x_{df} \geq T_\alpha$, a buy signal is generated, and when $x_{df} < T_\alpha$, a sell signal is generated. The dynamic threshold module optimizes the threshold T_α for a given stock from its historical data.

The dynamic threshold module optimizes the thresholds T_α using the particle swarm optimization algorithm. The formula of PSO is given as

$$v_t^{k+1} = \omega v_t^k + c_1 \times \text{rand}_1 \times (p_t^k - x_t^k) \\ + c_2 \times \text{rand}_2 \times (g_t^k - x_t^k) \quad (40)$$

$$x_t^{k+1} = x_t^k + v_t^{k+1} \quad (41)$$

where k is the iteration number, c_1 and c_2 are the learning factors and positive constants, rand_1 and rand_2 are random numbers between the range $[0, 1]$, ω is the inertia weight and is within $[0, 1]$. In the equation, ωv_t^k represents the influence

of the previous velocity of a particle on its current velocity. $c_1 \times \text{rand}_1 \times (p_t^k - x_t^k)$ represents the personal experience. $c_2 \times \text{rand}_2 \times (g_t^k - x_t^k)$ represents the collaborative effect of the particles and it always pulls the particle to the global best solution that the swarm has found so far.

At each generation, the velocity of each particle is calculated according to (40), and the position is updated by using (41). Each time, the better position is stored for the next generation, if a particle arrives at a promising new position, all the other particles will move closer to it. This procedure is repeated until satisfactory solution is found or the predefined number of iterations is met. In our method, c_1 and c_2 are set to 1.495, and the maximum number of iterations of the algorithm is set to 500.

The accuracy of forecasting trading points in the training set is used to construct the objective function given by

$$\text{Acc} = \frac{n_b(T_\alpha) + n_s(T_\alpha)}{n_{\text{total}}(T_\alpha)} \quad (42)$$

where n_b represents the number of correct predicted buy points, n_s denotes the number of correct predicted sell points, n_{total} denotes the total number of trading points, Acc represents the prediction accuracy. In the training set, the correct predicted buy points and sell points depend on the closing price of the following trading days. Therefore, ACP_i , CP_i in (24) are applied to determine the buy and sell points. The correct predicted buy points and sell points are defined as follows: If $\text{CP}_i < \text{ACP}_i$, then it is a buy points. If $\text{CP}_i > \text{ACP}_i$, then it is a sell points.

III. EXPERIMENTS AND RESULTS

In this section, we perform experiments to assess the performance of our method by making comparisons with six existing methods:

- 1) buy-and-hold (BAH) [26];
- 2) GP [38];
- 3) PLR-BPN [19];
- 4) PLR-WSVM [20];
- 5) BIC-K-NN [26];
- 6) an adaptive neuro-fuzzy inference system with reinforcement learning (ANFIS-RL) [39].

As stated earlier, the two parameters (i.e., T_{hc} and T_{msr}) determine the quality and quantity of the biclusters, and their optimal values are different for different stock. The range of T_{hc} is [0.001, 0.004] and the range of T_{msr} is [0.006, 0.02], and their optimal values are learned by grid search using the corresponding training data of each stock during the training phase.

The experimental data for the comparative study with the first five methods is based on two datasets. Table V shows the training data and test data in Dataset 1. Dataset 1 includes two stocks (Apple and BA) from U.S. and two stocks (AUO and UMC) from Taiwan. In PLR-BPN [19], the training data of the American stocks is selected from 2008/01/02 to 2008/12/30 while the test data are selected from 2009/01/02 to 2009/06/30. For each Taiwan stock, the training data are selected from 2006/01/02 to 2008/10/14 and the test data are selected from 2008/10/15 to 2009/04/09. Table VI shows the training data and test data

TABLE V
TRAINING DATA AND TEST DATA IN DATASET 1

Stock	Training period	Test period	Bicluster thresholds*
Apple	2008/01/02 - 2008/12/30	2009/01/02 - 2009/06/30	$T_{hc}=0.0010$ $T_{msr}=0.0060$
BA	2008/01/02 - 2008/12/30	2009/01/02 - 2009/06/30	$T_{hc}=0.0016$ $T_{msr}=0.0064$
AUO	2006/01/02 - 2008/10/14	2008/10/15 - 2009/04/09	$T_{hc}=0.0010$ $T_{msr}=0.0060$
UMC	2006/01/02 - 2008/10/14	2008/10/15 - 2009/04/09	$T_{hc}=0.0010$ $T_{msr}=0.0059$

*these parameters are determined by grid search

TABLE VI
TRAINING DATA AND TEST DATA IN DATASET 2

Stock		Training period	Test period	Bicluster thresholds*
Downtrend	600197	2010/01/04 – 2010/11/18	2010/11/19 – 2011/08/18	T_{hc} =0.0019 T_{msr} =0.0094
	600211	2010/01/04 – 2010/11/18	2010/11/19 – 2011/08/18	T_{hc} =0.0031 T_{msr} =0.0104
	600488	2010/01/04 – 2010/11/18	2010/11/19 – 2011/08/18	T_{hc} =0.0024 T_{msr} =0.0120
Steady trend	600066	2010/01/04 – 2010/11/18	2010/11/19 – 2011/08/18	T_{hc} =0.0027 T_{msr} =0.010
	600697	2010/01/04 – 2010/11/18	2010/11/19 – 2011/08/18	T_{hc} =0.0029 T_{msr} =0.0157
	600881	2010/01/04 – 2010/11/18	2010/11/19 – 2011/08/18	T_{hc} =0.0010 T_{msr} =0.0060
Uptrend	600051	2010/01/04 – 2010/11/18	2010/11/19 – 2011/08/18	T_{hc} =0.0014 T_{msr} =0.0074
	600107	2010/01/04 – 2010/11/18	2010/11/19 – 2011/08/18	T_{hc} =0.0028 T_{msr} =0.0081
	600167	2010/01/04 – 2010/11/18	2010/11/19 – 2011/08/18	T_{hc} =0.0037 T_{msr} =0.0099

*these parameters are determined by grid search

in Dataset 2. Dataset 2 includes nine stocks from the Shanghai Stock Exchange. In order to assess the performance of our method under different stock market condition, the nine stocks are divided into three types: downtrend (Code: 600197, 600211, 600488), steady trend (Code: 600066, 600697, 600881), and uptrend (Code: 600051, 600107, 600167). The test data are from 2010/11/19 to 2011/08/18, and the total number of days in the test data are about 200. For each stock, the training data starts from 2010/01/04 to 2010/11/18. For comparison with ANFIS-RL, we run our algorithm over the five stocks/indexes used in [39] (see Table XI), where we directly compare with the results reported there.

Profit maximization is the most important objective for stock forecasting. We use the profit ratio as the performance measure

$$\text{profit} = T \times \prod_{k=1}^n \left[1 + \frac{(1-t) \times \text{SP}_k - \text{BP}_k}{\text{BP}_k} \right] \quad (43)$$

$$\text{PR} = \left[\frac{\text{profit} - T}{T} \right] \times 100\% \quad (44)$$

where T represents the invested capital, t denotes the transaction fee, SP_k represents the sell price, BP_k denotes the buy price, profit denotes the total profit, and PR is the profit ratio.

TABLE VII
COMPARISON BETWEEN BM-FM AND OTHER METHODS IN DATASET 1

Stock	GP		PLR-BPN		BIC-K-NN		Our method	
	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times
Apple	10.20	10	61.28	13	37.60	8.0	50.13	20.0
BA	15.38	14	38.03	11	5.48	26.0	48.91	26.0
AUO	13.02	6	10.42	14	-6.83	24.0	17.42	18.0
UMC	5.41	6	63.13	10	34.13	16.0	72.25	38.0
Average	11.00	9	43.22	12	17.59	18.5	47.18	25.5

TABLE VIII
COMPARISON AMONG BAH, PLR-WSVM, BIC-K-NN AND BM-FM ON DOWNTREND

Stock	BAH		PLR-WSVM		BIC-K-NN		Our method	
	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times
600197	-24.05	2	11.29	4.00	6.72	24.00	27.83	20.00
600211	-13.20	2	5.38	18.00	20.07	24.00	26.40	24.00
600488	-27.53	2	-13.55	4.00	-4.02	8.00	6.06	8.00
Average	-21.59	2	8.34	8.67	8.92	18.67	20.09	17.33

TABLE IX
COMPARISON AMONG BAH, PLR-WSVM, BIC-K-NN, AND BM-FM ON STEADY TREND

Stock	BAH		PLR-WSVM		BIC-K-NN		Our method	
	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times
600066	-9.26	2	-3.88	4	13.39	22	15.36	18.00
600697	-3.82	2	4.30	18	17.75	18	23.19	6.00
600881	1.90	2	10.72	14	21.51	32	38.01	32.00
Average	-3.73	2	3.71	12	17.55	24	25.52	18.67

TABLE X
COMPARISON AMONG BAH, PLR-WSVM, BIC-K-NN AND BM-FM ON UPTREND

Stock	BAH		PLR-WSVM		BIC-K-NN		Our method	
	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times	Profit ratio (%)	Trading times
600051	16.43	2	21.78	16.00	35.92	24	40.81	22.0
600107	16.27	2	15.33	13.00	10.63	20	23.32	20.0
600167	19.47	2	13.48	9.00	37.75	34	54.65	14.0
Average	17.39	2	16.86	12.66	28.10	26	39.59	18.7

The algorithm is programmed with Qt Creator. The performance of our method is presented in Table VII to Table XI. The comparative results between BM-FM and other methods for Dataset 1 are listed in Table VII. Tables VIII–X show the results in different stock market condition (i.e., downtrend, steady trend, and uptrend). As can be seen, BM-FM shows good

prediction performance in comparison with the other methods in the testing period.

As shown in Table VII, the results of our method are better than GP and BIC-K-NN for all stocks. The profit ratio of our method is higher than PLR-BPN in three stocks (i.e., BA, AUO, and UMC). The average profit ratio of our method is 47.18%,

TABLE XI
COMPARISON WITH ANFIS-RL

Stock/Index	Profit ratio (%)		Bicluster thresholds*	Difference	Rank
	ANFIS-RL	Our method			
IBM	212.76	235.88	$T_{hc}=0.001, T_{msr}=0.006$	+23.12	2
Walmart	110.69	238.19	$T_{hc}=0.001, T_{msr}=0.005$	+127.50	6
Citigroup	131.68	105.38	$T_{hc}=0.001, T_{msr}=0.006$	-26.30	1
Dow Jones	195.91	216.29	$T_{hc}=0.001, T_{msr}=0.006$	+57.19	5
S&P	178.31	230.96	$T_{hc}=0.001, T_{msr}=0.005$	+52.65	4
Average	165.87	205.34		+39.47	3

*these parameters are determined by grid search

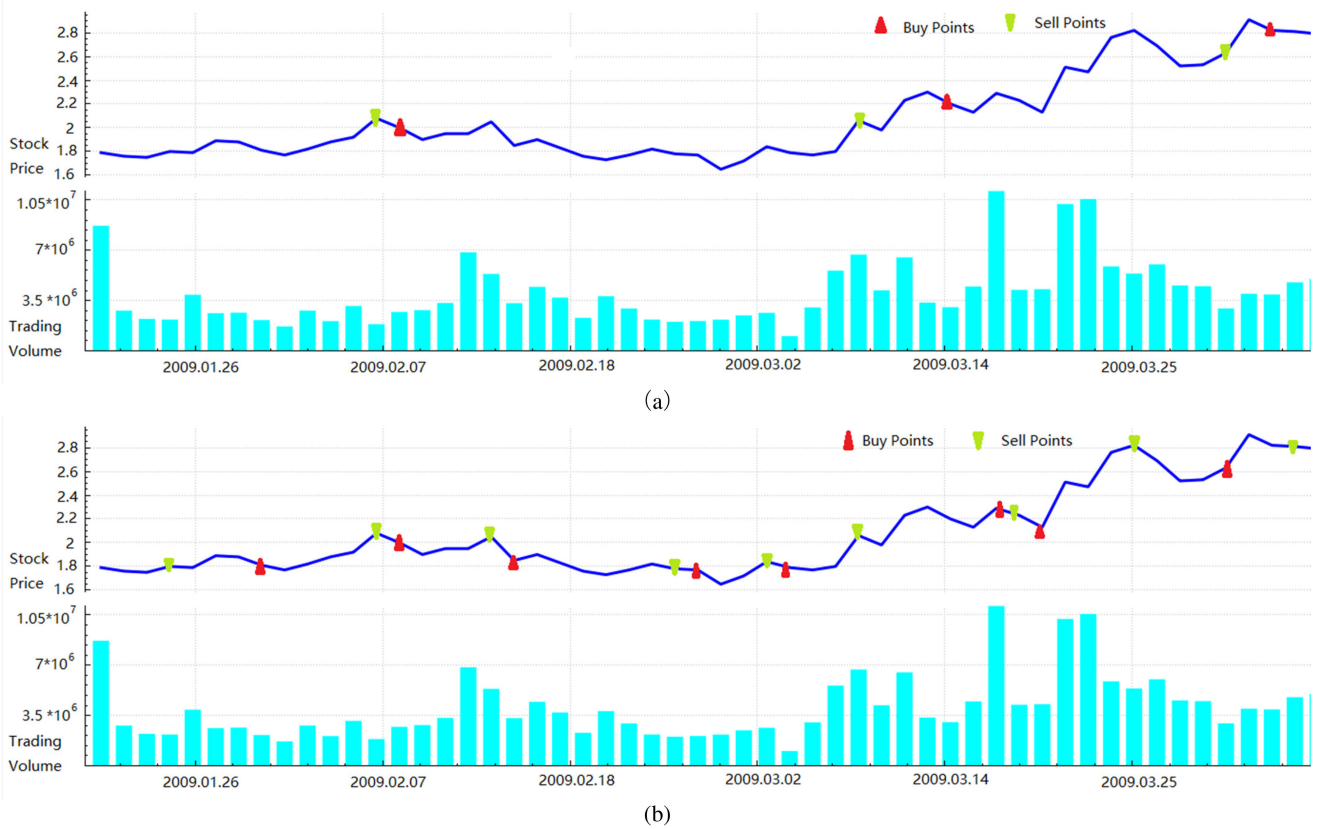


Fig. 7. Trading points for UMC from 2009/01/21 to 2009/04/09 obtained by (a) BIC-K-NN and (b) BM-FM.

which is significantly higher than GP (11.00%) and BIC-K-NN (17.59%). It is also slightly higher than PLR-BPN (43.22%).

As shown in Table VIII, the performance of BAH is poor on the downtrend where the average profit ratio is negative. But the average performance of our method gains 20.09% and is better than all the compared methods. It suggests that BM-FM has stable prediction performance even in the downtrend situation. Table IX shows that our method outperforms all the other methods on the steady trend. Table X shows that all methods gain positive average earning on the uptrend but our method obtains the highest profit ratio. On the whole, our method has superior performance under different stock market conditions.

The comparison with ANFIS-RL [39] is shown in Table XI. From Table XI, we can see that our method obtains the highest profit ratios in most cases. According to the Wilcoxon ranked sign test [39], the test statistics computed from Table XI is -1.979 . It shows that our proposed method is better than ANFIS-RL at a confidence level of 0.05.

Figs. 7–9 show three stocks (i.e., UMC, 600211, 600167) that are marked with buy and sell trading points obtained from BIC-K-NN and BM-FM for the test period. Comparison between BIC-K-NN and BM-FM in the test period reveals that the results of BIC-K-NN did not perform well sometimes. Our method shows better performance in the comparative experiments.

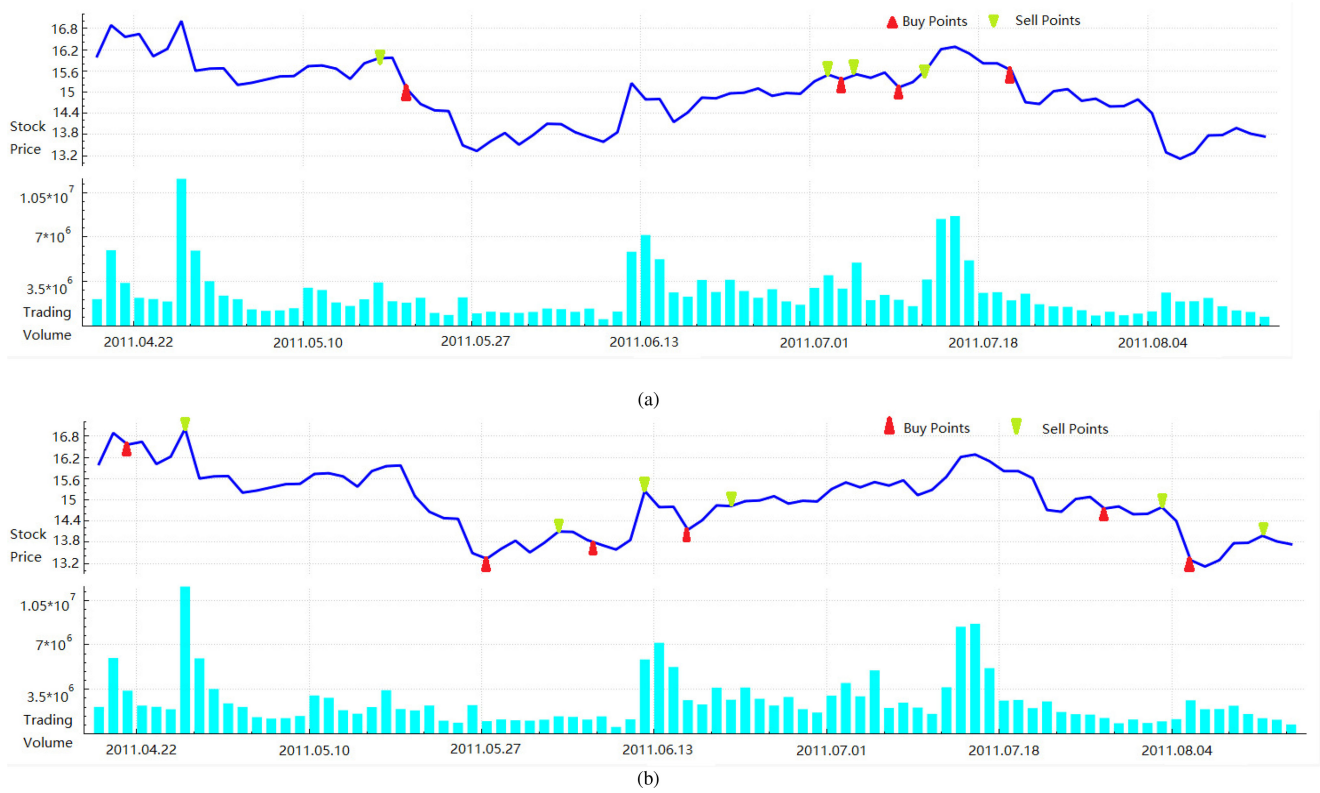


Fig. 8. Trading points for 600211 from 2011/04/19 to 2011/08/17 obtained by (a) BIC-K-NN and (b) BM-FM.

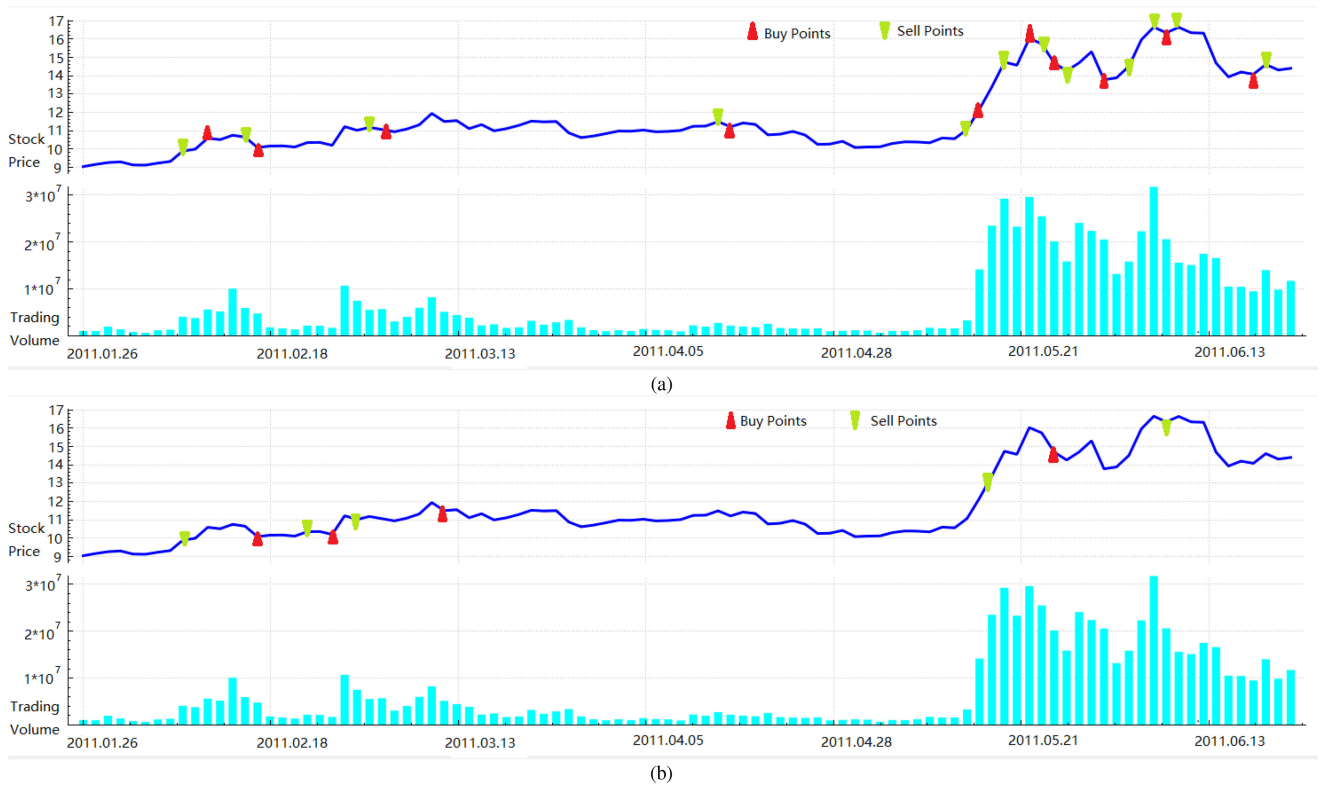


Fig. 9. Trading points for 600167 from 2011/01/26 to 2011/06/25 obtained by (a) BIC-K-NN and (b) BM-FM.

Choosing the right combination of technical indicator(s) for a particular trading condition is a challenging unsolved problem in automatic trading systems. In our algorithm, biclustering is used to mine the trading rules from the technical indicators computed from past stock trading data. Different time periods are used for the technical indicators where applicable, and the time period captures the temporal information in the trading data. The use of biclustering allows us to automatically determine the subset of relevant indicators for each trading point signal and for each stock. Currently, there is no principled way to do this in an automatic trading system, and our system is the first to do so.

Two common fuzzy inference system (FIS) are the Mamdani FIS and the Takagi–Sugeno (TS) FIS. In this paper, we have adopted the Mamdani FIS. Mamdani FIS has the advantage that its output is also a fuzzy set, and is therefore more easily interpretable than TS FIS, whose output is a linear function. In our automatic stock trading system, the future return rate is a linguistic variable (i.e., return is quantified in terms of “negative large”, “negative small”, “positive small”, “positive large”) and can be naturally represented as fuzzy membership functions. Mamdani FIS is, therefore, a natural choice in this paper.

In our FIS, we use a dynamic threshold module to determine the threshold T_α to buy/sell given the defuzzified output x_{df} given by (39). The threshold is determined by the particle swarm optimization given the closing price of the following day. This mechanism gives our FIS the ability to adapting to changing market condition to some extent. For example, we could envisage a scenario where some market event leads to risk aversion, and what would normally be a “buy” would now be “hold” or even a “sell”. A dynamic threshold updated continuously according to current market condition can provide a way to capture this market dynamic.

In classification systems, noise in the label (mislabelled data) or in the attributes (for example, due to missing value, sensor errors, measurement noise) can adversely affect the performance [41]. In order to evaluate, the impact of noise on the technical indicators for the profit ratio, two experiments are performed with the Apple stock dataset according to [41] (each experiment is done with ten repetitions). Specifically, a technical indicator x is corrupted with noise at a noise level of ϑ by randomly selecting $9 \times 100\%$ of x and corrupt each of the value by replacing them with a uniformly sampled value from the range $[v - v * \vartheta, v + v * \vartheta]$, where v is the original normalized uncorrupted value of x .

In the first experiment, we added random noise of different level (i.e., ϑ ranges from 0.0 to 0.2) to all the technical indicators to investigate the overall impact of attribute noise. In Fig. 10, the blue curve is the profit ratio without attribute noise, and the red curve is the profit ratio with attribute noise in all the technical indicators. From Fig. 10, we see that the profit ratio will degrade with the increase in noise level, but overall the degradation is still moderate. Note that a noise level of 0.2 is quite severe as it means the value v is corrupted by up to $\pm 20\%$ of its original value.

In the second experiment, we want to investigate which technical indicator is more sensitive to attribute noise. Only one of the technical indicators is corrupted with attribute noise and the

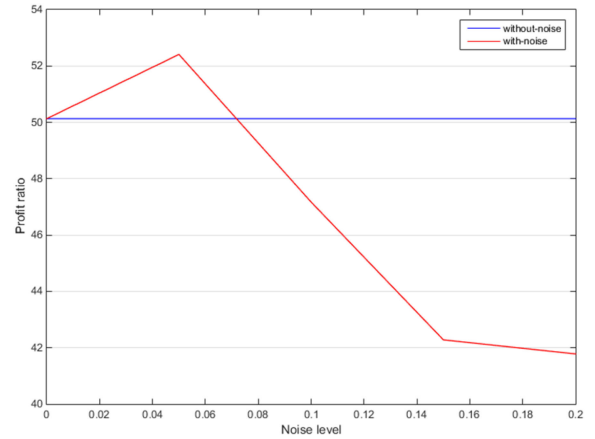


Fig. 10. Overall impact of attribute noise to profit ratio.

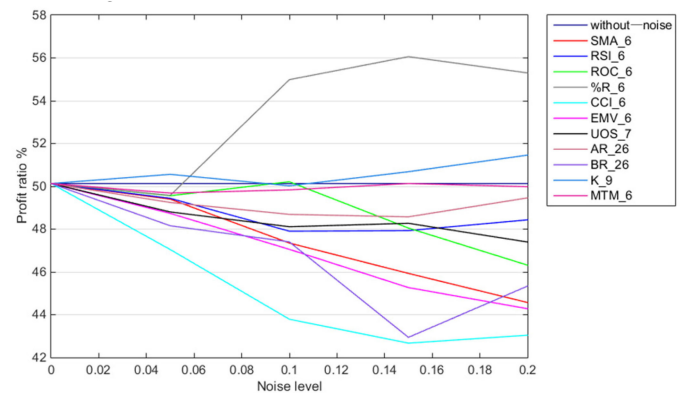


Fig. 11. Impact of noise on several technical indicators (SMA_6, RSI_6, ROC_6, %R_6, CCI_6, EMV_6, UOS_7, AR_26, BR_26, MTM_6) on the profit ratio.

experimental results of several indicators are shown in Fig. 11 (see supplementary file for complete results). We can see that some attribute noise will increase the profit ratio, like %R_6 and K_9, but some attribute noise will decrease the profit ratio, like CCI_6, SMA_6, and EMV_6. Overall, the degradation of the profit ratio is still moderate with the increasing level of noise.

Although there are other ways to handle the attribute noise problem [42], [43], we see that our algorithm is not very sensitive to attribute noise due to the use of the averaging operator in the computation of the vector of average technical indicators from the bicluster by (27). The averaging operation smoothed out noise and inconsistencies in the values, thereby ensuring a more consistent set of fuzzy rules to be obtained.

The training phase of our algorithm cannot be run in real time due to the expensive biclustering and parameter optimization process. However, once the model is trained and fixed, inference is straightforward and can be done in real time.

IV. CONCLUSION

This paper proposed a novel intelligent trading method (BM-FM) combining a bicluster mining algorithm and a fuzzy inference system to automatically discover trading rules from a

combination of technical indicators in the financial markets. In this paper, 13 technical indicators (SMA, RSI, %R, ROC, CCI, EMV, UOS, BR, AR, K, D, J, and MTM) with different time period parameters, and future returns are used to form the data matrix. Our algorithm first detects biclusters with constant columns from the data matrix. It then converts the discovered biclusters into fuzzy trading rules, which are grouped into buy and sell signals. Finally, a dynamic threshold value learned using PSO is used to determine the trading action for a stock in a trading day. In contrast to many fuzzy inference systems where the fuzzy rules are defined by experts, the fuzzy rules in our system are mined from the trading data automatically. The proposed algorithm is tested on three datasets and compared with six existing methods. The results demonstrate that our method significantly achieve better performance than its counterparts.

In a fuzzy inference system, the quality of the fuzzy rules determines the accuracy of the prediction. Therefore, fuzzy rule selection could improve performance [44]. In future paper, we plan to perform rule selection to select a subset of optimal fuzzy rules to achieve better prediction performance. Another direction is to incorporate some kind of online learning mechanism to handle the change in market dynamic. Although our adaptive threshold module can provide some adaptation, the underlying inference model is still static. An online learning mechanism that could explicitly handle concept drifts would allow truly dynamic adaptation to the underlying distribution [45].

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